

WORK, ENERGY AND POWER

(1) WORK

Work is said to be done when a force acting on a body displaces the body in the direction of the force.

$$W = \vec{F} \cdot \vec{S} = Fs \cos \theta$$

SI Unit → Joule (J)
[1 J = 1 N·m]

Where, $F \rightarrow$ Force applied (N)

$S \rightarrow$ Displacement (m)

$\theta \rightarrow$ Angle between $\vec{F}$ and $\vec{S}$

Nature of Work

- Scalar quantity
- Can be +ve, -ve or zero
- Depends on magnitude of force, displacement and angle between them.

Case Study (Angle θ)

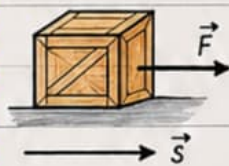
• $\theta = 0^\circ \rightarrow W = Fs$ (+ve)

• $\theta = 90^\circ \rightarrow W = 0$ (Zero)

• $\theta = 180^\circ \rightarrow W = -Fs$ (-ve)

• $\theta > 90^\circ \rightarrow W$ is negative

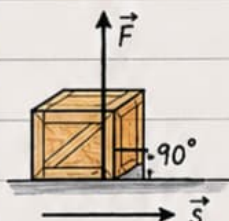
$\theta = 0^\circ$



$W = +Fs$

Positive Work

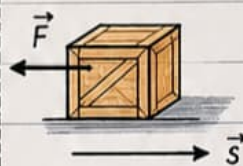
$\theta = 90^\circ$



$W = 0$

Zero Work

$\theta = 180^\circ$



$W = -Fs$

Negative Work

Examples

- ① Pushing a box in forward direction $\rightarrow$ Positive work
- ② Lifting a bag upward $\rightarrow$ Positive work
- ③ Holding a bag without moving $\rightarrow$ Zero work
- ④ Friction acting on a moving object $\rightarrow$ Negative work

(2) KINETIC ENERGY (K.E.)

The energy possessed by a body due to its motion.

$$K.E. = \frac{1}{2} mv^2$$

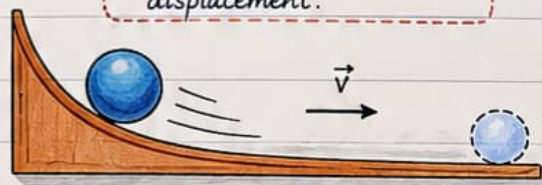
Where, $m \rightarrow$ mass (kg)
 $v \rightarrow$ velocity (m/s)

SI Unit $\rightarrow$ Joule (J)

K.E. is always positive.

Important Points

- Work is done only when displacement occurs.
- If either $F = 0$ or $S = 0$ then $W = 0$.
- Work depends on the component of force in the direction of displacement.



As speed increases,
K.E. increases.

③ POTENTIAL ENERGY (P.E.) 🌿

The energy possessed by a body due to its position or configuration.

• Gravitational Potential Energy (U)

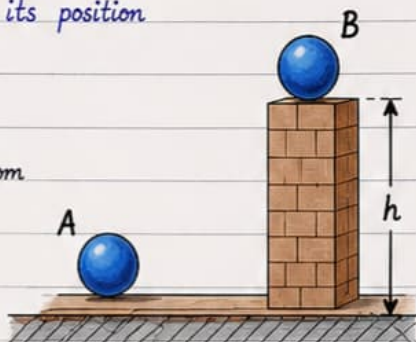
When a body is raised to a height 'h' from the ground,

$$U = mgh$$

Where, $m \rightarrow$ mass (kg)

$g \rightarrow$ acceleration due to gravity ($\approx 9.8 \text{ m/s}^2$)

$h \rightarrow$ height (m)



Key Points

- P.E. depends on height and mass.
- Unit $\rightarrow$ Joule (J)
- P.E. can be +, - or zero.
- At $h = 0$, $U = 0$.

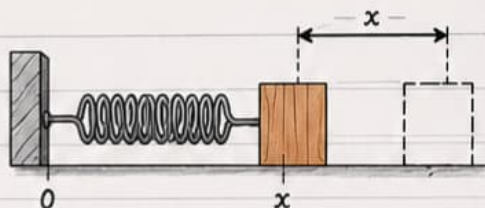
• Elastic Potential Energy

When a spring is stretched or compressed, it stores energy.

$$U_s = \frac{1}{2} kx^2$$

Where, $k \rightarrow$ spring constant (N/m)

$x \rightarrow$ extension or compression (m)



Mechanical Energy (M.E.)

The total mechanical energy of a body is the sum of its kinetic energy and potential energy.

$$M.E. = K.E. + P.E.$$

Unit $\rightarrow$ Joule (J)

"M.E. is conserved if only conservative forces act." ☆

④ WORK - ENERGY THEOREM 🌿

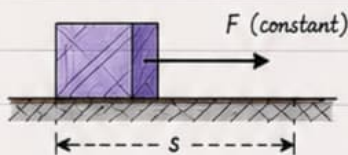
The net work done on a body is equal to the change in its kinetic energy.

$$W_{\text{net}} = \Delta K.E. = K.E._f - K.E._i$$

Where, $K.E._i \rightarrow$ initial kinetic energy

$K.E._f \rightarrow$ final kinetic energy

$W_{\text{net}} \rightarrow$ net work done



For constant force along displacement,

$$W = Fs \cos \theta$$

If $\theta = 0^\circ$ (force in direction of motion) $\rightarrow W = Fs$

⑤ POWER (P) 🌿

Power is the rate at which work is done.

Average Power

$$P_{\text{avg}} = \frac{W}{t}$$

Instantaneous Power

$$P = \frac{dW}{dt}$$

(In calculus form)

Where,

$W \rightarrow$ work done (J)

$t \rightarrow$ time taken (s)

$P \rightarrow$ power (Watt)

SI Unit

1 Watt (W) = 1 Joule per second

[1 W = 1 J/s]

⇌ Examples ⇌

- ① A body of mass 2 kg is raised to a height of 5 m.
Find its P.E. ($g = 10 \text{ m/s}^2$)

Sol.

$$U = mgh = 2 \times 10 \times 5$$

$$U = 100 \text{ J}$$

- ② A body of mass 3 kg is moving with a velocity of 4 m/s.
Find its K.E.

Sol.

$$K.E. = \frac{1}{2} mv^2 = \frac{1}{2} \times 3 \times 4^2$$

$$K.E. = 24 \text{ J}$$

- ③ A force of 20 N acts on a body in the direction of motion through a displacement of 10 m.
Find the work done.

Sol.

$$W = Fs = 20 \times 10$$

$$W = 200 \text{ J}$$

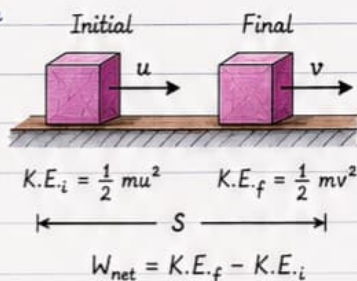


6 KINETIC ENERGY THEOREM

The net work done on a body is equal to the change in its kinetic energy.

$$W_{\text{net}} = \Delta K.E. = K.E._f - K.E._i$$

This is another form of Work-Energy Theorem.



Key Points

- If $W_{\text{net}} > 0$ then K.E. increases.
- If $W_{\text{net}} < 0$ then K.E. decreases.
- If $W_{\text{net}} = 0$ then K.E. remains constant.

7 CONSERVATION OF MECHANICAL ENERGY

If only conservative forces act on a body, then the total mechanical energy remains constant.

$$M.E._i = M.E._f$$

or

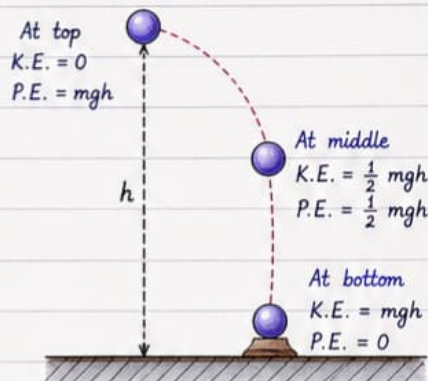
$$K.E._i + P.E._i = K.E._f + P.E._f$$

Example :

A body falling freely under gravity (no air resistance)

Top : P.E. maximum, K.E. = 0

Bottom : K.E. maximum, P.E. = 0



8 POWER AND ITS UNIT

Power is the rate at which work is done.

$$P = \frac{W}{t}$$

Where,

$W \rightarrow$ work done (J)

$t \rightarrow$ time taken (s)

$P \rightarrow$ power (Watt)

SI Unit $\rightarrow$ Watt (W)

1 Watt = 1 Joule / second

[1 W = 1 J/s]

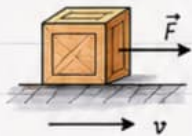
Power is a scalar quantity. It can be +ve, -ve or zero.

9 RELATION BETWEEN FORCE, VELOCITY AND POWER

When a constant force F acts on a body moving with velocity v in the direction of force.

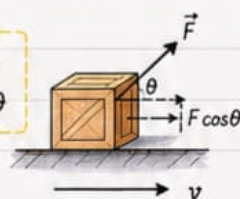
$$P = \frac{W}{t} = \frac{Fs}{t} = F \left(\frac{s}{t} \right) = Fv$$

$$P = Fv \quad (\text{when } \theta = 0^\circ)$$



In general,

$$P = Fv \cos \theta$$



Key Points

- If $0^\circ \leq \theta < 90^\circ \rightarrow$ Power is +ve
- If $\theta = 90^\circ \rightarrow$ Power is zero
- If $90^\circ < \theta \leq 180^\circ \rightarrow$ Power is -ve

10 IMPORTANT CASES

① Lifting an object vertically upward



$$W = +mgh$$

(Positive work)

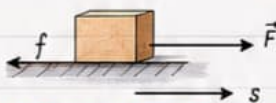
② Lowering an object vertically downward



$$W = -mgh$$

(Negative work)

③ Pushing a box on rough horizontal surface



$$W = (F \cos \theta - f)s$$

If $F \cos \theta > f \rightarrow$ +ve work

If $F \cos \theta < f \rightarrow$ -ve work

④ Tension doing work on a hanging mass



$$W = +mgh$$

(Positive work)

Quick Revision

- Work is done by a force.
- $W = Fs \cos \theta$
- Unit $\rightarrow$ Joule (J)

- $K.E. = \frac{1}{2} mv^2$
- P.E. (gravity) = mgh
- P.E. (spring) = $\frac{1}{2} kx^2$

- $W_{\text{net}} = \Delta K.E.$
- M.E. conserved (only conservative forces)

- Power $P = \frac{W}{t}$
- $P = Fv \cos \theta$
- Unit $\rightarrow$ Watt (W)

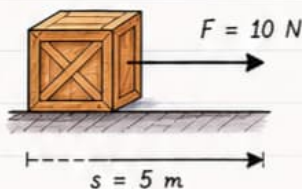
Worked Examples 🌿

Example 1:

A force of 10 N acts on a body in the direction of motion and produces a displacement of 5 m. Find the work done.

Solution:

$$\begin{aligned} W &= Fs \cos \theta \\ &= 10 \times 5 \times \cos 0^\circ \\ &= 50 \times 1 \\ W &= 50 \text{ J} \end{aligned}$$



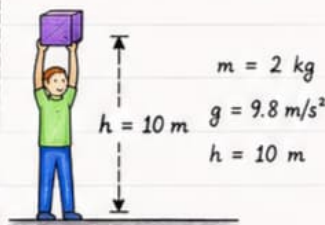
Since force is in the direction of motion,
 $\theta = 0^\circ \Rightarrow \cos \theta = 1$

Example 2:

A body of mass 2 kg is raised vertically to a height of 10 m. Find the work done against gravity.

Solution:

$$\begin{aligned} W &= mgh \\ &= 2 \times 9.8 \times 10 \\ W &= 196 \text{ J} \end{aligned}$$

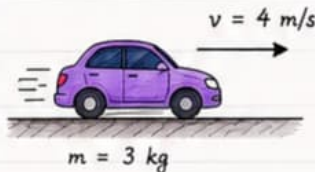


Example 3:

A body of mass 3 kg is moving with a velocity of 4 m/s. Find its kinetic energy.

Solution:

$$\begin{aligned} \text{K.E.} &= \frac{1}{2} mv^2 \\ &= \frac{1}{2} \times 3 \times (4)^2 \\ &= \frac{1}{2} \times 3 \times 16 \\ \text{K.E.} &= 24 \text{ J} \end{aligned}$$



Chapter Summary ☆

☆ Work is done when a force causes displacement. $W = Fs \cos \theta$

☆ Work can be positive, negative or zero depending on angle between force and displacement.

☆ Kinetic Energy is the energy of a body due to its motion.
 $\text{K.E.} = \frac{1}{2} mv^2$

☆ Potential Energy is the energy stored due to position or configuration.

- Gravitational P.E. = mgh
- Elastic P.E. = $\frac{1}{2} kx^2$

☆ Mechanical Energy is the sum of K.E. and P.E.

$$\text{M.E.} = \text{K.E.} + \text{P.E.}$$

☆ According to Work-Energy Theorem,

$$\begin{aligned} W_{\text{net}} &= \Delta \text{K.E.} \\ &= \text{K.E.}_f - \text{K.E.}_i \end{aligned}$$

☆ Power is the rate of doing work.

$$P = \frac{W}{t}$$

☆ Unit of work and energy is Joule (J).

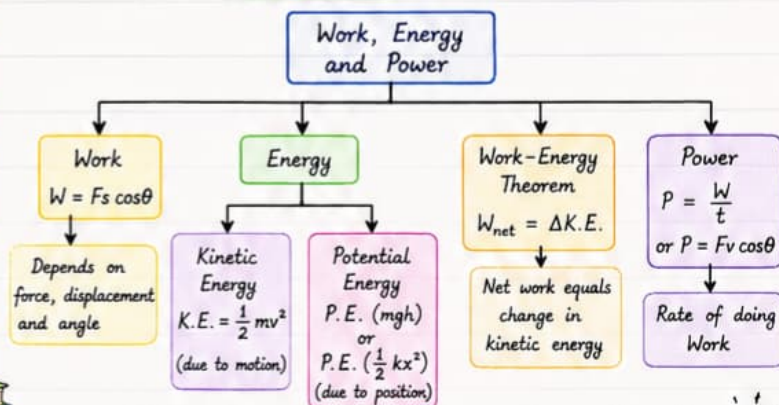
$$1 \text{ J} = 1 \text{ N} \cdot \text{m}$$

Key Formulas ☆

- Work done : $W = Fs \cos \theta$
- Kinetic Energy : $\text{K.E.} = \frac{1}{2} mv^2$
- Gravitational P.E. : $\text{P.E.} = mgh$
- Elastic P.E. : $\text{P.E.} = \frac{1}{2} kx^2$
- Mechanical Energy : $\text{M.E.} = \text{K.E.} + \text{P.E.}$
- Work-Energy Theorem : $W_{\text{net}} = \Delta \text{K.E.}$
 $= \text{K.E.}_f - \text{K.E.}_i$
- Power : $P = \frac{W}{t}$ or $P = Fv \cos \theta$
- Average Power : $P_{\text{avg}} = \frac{W}{t}$
- Instantaneous Power : $P = \frac{dW}{dt}$



Revision Map 🌿



Energy can change from one form to another but the total energy of an isolated system remains constant. ☆



Points to Remember ♥

- No displacement $\Rightarrow$ No work.
- Work depends on both magnitude and direction.
- Kinetic Energy is always positive.
- Potential Energy can be positive, negative or zero.
- Mechanical Energy is conserved if only conservative forces act.

Understand the concept
 Deeply, Practice Regularly and
 You will **Master Physics!** ☆

